# TraceCase

Bug Reproduction Agent / Developer & Enterprise AI

## A vague bug report is not enough to reproduce a defect.

Missing steps and environment details force developers to repeat investigation work.

**PS requirement** Reproduce twice and provide recording, console logs and network evidence.

# The solution

A reviewed browser investigation with a traceable developer handoff

## Natural-language intake

Select a target and describe the symptom.
Optional context helps, but users do not need to write selectors.

## Human control

Review the proposed steps before execution.
Destructive actions require an authorized sandbox.

## Browser observations

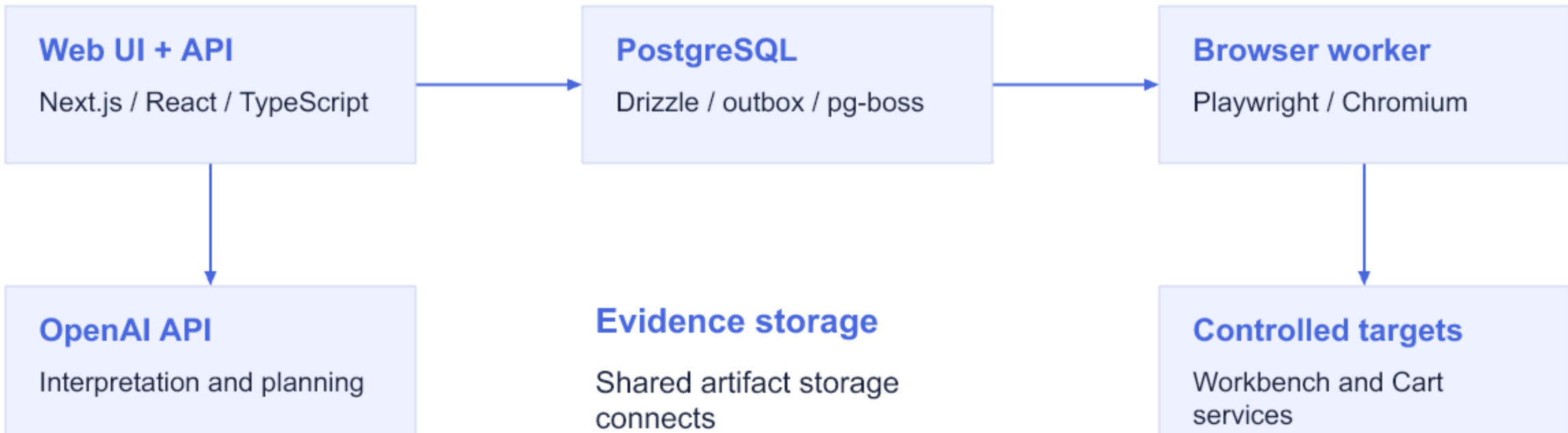
Playwright performs the approved actions.
Two attempts compare behavior under a restored baseline.

## Results and evidence

Inspect actions, verdict rationale and evidence availability.
Export the investigation as PDF or JSON.

# System architecture

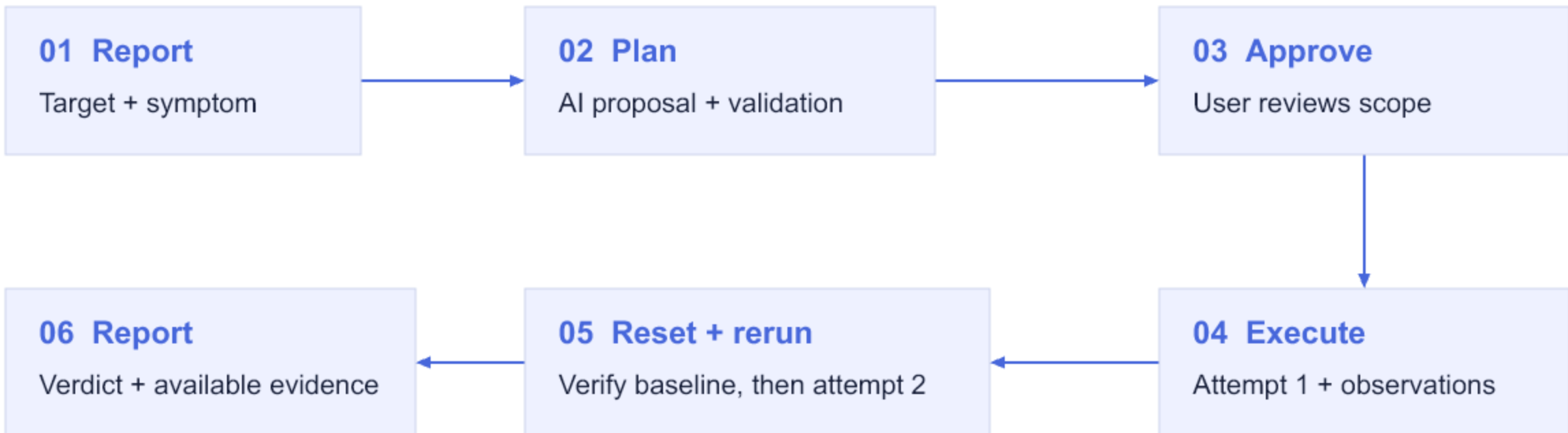
Separate Render services connect through durable workflow state




Approval, tool validation and deterministic verification constrain AI proposals.

# Investigation workflow

The user reviews the plan before the worker takes action




Clarification resolves essential ambiguity. Errors and missing evidence remain visible.

# Verification methodology

Four official acceptance scenarios define the demonstration

| | | |
|---|---|---|
| **TC1** | **Reproduced twice** | Matching observed failures, verified reset and recording / console / network evidence. |
| **TC2** | **Cannot reproduce** | Valid attempted steps and environment, with an honest Not Reproduced outcome. |
| **TC3** | **Destructive action** | Deletion stays within a verified sandbox and requires explicit approval. |
| **TC4** | **Token redaction** | A fake token canary is absent from shared logs, traces and exported reports. |

**Release gate: verify all four in the live browser. This deck does not assert a four-case pass.**

# Impact and scalability

Designed to reduce investigation effort without hiding uncertainty

## Developer value

Turn an informal report into reviewable steps and observed results.
Keep the reproduction context with the handoff.

## Efficient AI use

Use AI for language and planning.
Keep authorization, arithmetic and verdict checks deterministic.

## Scaling path

Add workers as capacity grows.
Coordinate target access and isolate fixtures to avoid interference.

## Next validation

Measure completion rate, time to evidence and cost per investigation.
No measured improvement is claimed yet.

# References and project access

Project sources define the requirements, architecture and safety contract

## Source repository

https://github.com/sharancode3/TraceCase

## Requirements and design

docs/prd.md
docs/system-architecture.md
docs/verification-evidence-and-safety-specification.md

## Implementation sources

render.yaml package.json worker/ src/server/